

Poisson-Boltzmann equation

In this tutorial, we investigate how the ions in an aqueous solution organize close to a surface carrying a surface charge σ . We consider that there are several ionic species with charge $q_i = z_i q$ and concentration n_{i0} far from the charged surface, where i indicates the species, q is the elementary charge and the integer z_i is the valency.

1 Poisson-Boltzmann equation

1. What are the most abundant ions in sea water? What are the corresponding q_i and n_{i0} ? Write the electroneutrality condition in the general case.
2. We now assume that the concentrations may depend on the position $\mathbf{r}$ and we write $n_i(\mathbf{r})$. What is the charge density $\rho(\mathbf{r})$ at a given position? Use the Maxwell-Gauss equation or the Poisson equation to write a relation between the electrostatic potential $V(\mathbf{r})$ and the charge density.
3. What is the energy $U_i(\mathbf{r})$ of an ion with charge q_i located at the position $\mathbf{r}$ in the electrostatic potential $V(\mathbf{r})$. The Boltzmann distribution gives the density of ions of type i : $n_i(\mathbf{r}) = Z_i^{-1} \exp(-U_i(\mathbf{r})/k_B T)$, where Z_i is a normalization factor. How should the normalization factor be chosen so that the density far from the charged surface is n_{i0} ?
4. Combine the equations obtained in the two previous questions to eliminate the concentrations $n_i(\mathbf{r})$ and obtain an equation for the electrostatic potential $V(\mathbf{r})$. This is the Poisson-Boltzmann equation.

2 Debye-Hückel limit and Debye length

5. When is the argument of the exponential functions appearing in the Poisson-Boltzmann equation small? Assume that it is small and linearize the Poisson-Boltzmann equation by using $\exp(1+u) \simeq 1+u$; this is the Debye-Hückel limit. You need to use the electroneutrality condition.
6. Write the resulting equation under the form $\nabla^2 V(\mathbf{r}) = \ell_D^{-2} V(\mathbf{r})$, where ∇^2 is the Laplacian; this is the linearized Poisson-Boltzmann equation. The characteristic length ℓ_D is the Debye length. Write it explicitly and estimate its value in sea water and in physiological conditions.

We now assume that the charged surface is a plane located at $x = 0$, and that the solutions lies in the half-space $x > 0$. In this situation, the electrostatic potential depends only on the distance x to the plane and we write $V(x)$.

7. Write and solve the equation for $V(x)$ by taking the boundary condition $V(0) = V_s$.
8. From the solution obtained in the previous question, what is the electric field $\mathbf{E}$ close to the surface, at $x = 0^+$? From the Maxwell-Gauss equation, justify that, if the electric field is zero for $x < 0$, $\mathbf{E}(0^+) = (\sigma/\epsilon)\mathbf{e}_x$. Use this expression to relate the surface charge σ and the surface potential V_s . Discuss their dependence on the various parameters.
9. Integrate the ionic concentrations from $x = 0$ to $x \rightarrow \infty$ to get the electric charge close to the surface. How does it compare to σ ?
10. Why is this phenomenon called *screening*? Discuss the difference between a charged medium and a polarizable medium in the way they affect the electric field.

3 Beyond the Debye-Hückel limit

We now focus on a symmetric monovalent salt, meaning that there are two ionic species with $z_+ = 1$, $z_- = -1$, $n_{+0} = n_{-0} = n_s$.

11. Write the nonlinear Poisson-Boltzmann equation in this case.
12. Write the equation for the reduced potential $\psi(\mathbf{r}) = qV(\mathbf{r})/(k_B T)$ and identify the Debye length. What is the physical dimension of the reduced potential? With this quantity, what does the Debye-Hückel approximation correspond to?

We come back to the case of a planar charged surface at $x = 0$ facing a semi-infinite medium. The solution to ordinary differential equation $v''(x) = \sinh(v(x))$ is $\tanh(v(x)/4) = A \exp(-v(x))$, where A is a constant.

13. Solve for the potential $\psi(x)$ given its value at the surface, $\psi(0) = \psi_s$.
14. Given that $\tanh' = 1 - \tanh^2$, use the same argument as before to relate the surface potential ψ_s to the surface charge σ .
15. Introduce a reduced surface charge and identify the regimes of weak and large surface charges.
16. Check that you can recover the results of the previous section.